Harper Archambault

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https://harperarch.github.io/portfolio-site/

Graduate Computer Science student at the University of Central Florida with a strong foundation in software development, systems, and machine learning. Experienced in Python, C, Java, and JavaScript, with hands-on project work in scalable systems, networking, and AI-driven applications. Seeking internships and early-career software engineering roles.

Education

University of Central Florida **Orlando, FL**

Master of Science in Computer Science *Expected December 2026*

- GPA 3.86
- Research Pair Track: Machine Learning

Bachelor of Science in Computer Science *December 2025*

- Minor: Mathematics
- GPA 3.76
- Accelerated B.S. to M.S. Program

Academic Projects

Buyer Beware — AI-Driven Network Firewall

Senior Design Project, 2025

- Developed a Python-based real-time web scraping and network analysis system using mitmproxy to intercept and process HTTP/HTTPS traffic.
- Designed a Redis-based streaming pipeline to clean, deduplicate, and route data to AI models for content detection.
- Implemented scalable parsing and performance monitoring; deployed and validated the system on a Raspberry Pi router.
- Collaborated on interdisciplinary research integrating networking, security, and AI-driven text analysis.

Evaluating Prompt Injection Attacks in Large Language Models

NLP / Security Research Project, 2026

- Designed and implemented an evaluation framework to test prompt injection vulnerabilities across LLMs (GPT-4o, Claude Sonnet 4.5)
- Built a curated dataset of adversarial prompts across three attack categories: instruction override, role misdirection, and jailbreak chaining
- Developed an automated evaluation pipeline with attack success rate (ASR) metrics and manual validation of model outputs
- Analyzed effectiveness of system prompt hardening, reducing attack success rates by over 50% in vulnerable cases
- Conducted controlled experiments with deterministic decoding to ensure reproducibility across model-condition pairs

Tomasulo Algorithm Optimization Project

Advanced Computer Architecture Research Project, 2025

- Built an enhanced dynamic instruction scheduling simulator in Python using queueing-theory-based performance modeling.
- Modeled functional units as M/D/1 queues and applied SLSQP constrained optimization to minimize average instruction latency under fixed cost budgets.
- Implemented dynamic service-rate adaptation using EMA feedback, achieving 9–34% latency reduction across multiple workload profiles.

- Strengthened understanding of processor pipelines, out-of-order execution, and performance modeling.

Self-Supervised Representation Learning & PPO Hyperparameter Study
Machine Learning Research Projects, 2025

- Implemented a self-supervised visual representation learning pipeline on CIFAR-10 and evaluated transfer to SVHN using linear probes.
- Conducted a PPO hyperparameter sensitivity study in PyTorch, analyzing convergence, stability, and sample efficiency.
- Generated and analyzed reward curves across 5+ configurations to compare learning performance.

Computer Architecture Simulators
Computer Architecture Projects, 2024

- Implemented a cache simulator (LRU/FIFO, WB/WT) and a gshare branch predictor in C to study prediction accuracy and memory-hierarchy performance.
- Explored associativity, LRU/FIFO replacement, write-back vs write-through policies, and misprediction rate evaluation.

UCF Animal Tracker
Object-Oriented Software Development MERN Project, 2024

- Created a full-stack MERN web application enabling real-time wildlife reporting with GPS-based location tracking and data visualization.
- Designed secure REST APIs and integrated MongoDB Atlas for live updates and optimized query performance.
- Collaborated with team members on frontend development and proactively implemented key functionality to the application

Connexus
Object Oriented Software Development, 2024

- Developed a LAMP-stack contact management system with CRUD functionality and API integration in PHP and MySQL.
- Deployed the application on DigitalOcean, configuring virtual server networking and database security.

Technical Skills

- Programming Languages: Python, C, Java, JavaScript, TypeScript, PHP, HTML
- Networking & Systems: mitmproxy, Redis, iptables, Linux (Ubuntu), Docker
- AI & Data Processing: OpenAI API, Claude API, Pandas, NumPy, data cleaning & streaming pipelines
- Frontend: React, React Native, Tailwind CSS
- Backend & APIs: Node.js, Express
- Databases: MongoDB, MySQL
- Tools: GitHub, Postman, SwaggerHub, Visual Studio Code, Unity
- Concepts: Network Security, AI-Driven Automation, System Architecture, Object-Oriented Design, Agile Development

Professional Experience

Shift Manager — Plato's Closet *May 2025 – Present*

- Supervise daily operations and collaborate with a team of 10+ employees, ensuring efficient workflows and customer satisfaction in a high-volume retail environment.
- Manage POS systems and cash reconciliation balancing operational efficiency with customer engagement.
- Assist with developing and training new employees on store protocols, demonstrating leadership, accountability, and adaptability under pressure.